**Government budget:** The government budget is the amount of expenditure they project to spend on the economy.

Total government expenditure = G

Total government revenue = R

If G>R, the government is in a budget deficit

If G<R, then the government is in a surplus

If G=R, then the government is balanced

**Government borrowing:**

1.) Internal borrowing – borrowing money from inside the economy (for example, taking a loan from the central bank).

2.) External borrowing – borrowing money from outside the economy (for example, taking a loan from another country).

National debt: Total money borrowed by the government.

## Taxes

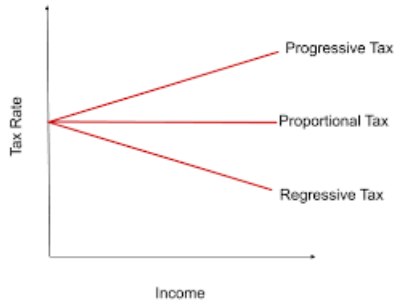
**3 different types of taxes:**

1.) Progressive taxes – When the proportion of the tax increases, as your income increases. For example, for income taxes, you have to pay a greater percentage of your income as you start to earn more.

2.) Regressive taxes – When you take a greater tax percentage from those who earn less. An example would be indirect taxes. This could be bad for an economy as it increases the gap in income distribution in your economy.

3.) Proportional tax – This is when the tax has a fixed rate regardless of your income. For example, VAT normally has a fixed percentage of tax.

**Tax rate diagram:**




Direct tax: Tax on income, for example corporate taxes.

Indirect tax: Taxes on expenditures like taxes on buying a cigarette.

**Pros of direct taxes:**

1.) It gives revenue to the government

2.) It seems fair because you're earning a lot so you can pay back a proportion of it

3.) They are based on your ability to pay

**Cons of direct taxes:**

1.) They discourage consumption so it decreases AD (resulting in a decrease in GDP)

2.) If income taxes are high, they can demotivate working and labor

3.) Corporate taxes can restrict companies from reinvesting in their company hence resulting in a decrease in AD

**Pros of indirect taxes:**

1.) They are flexible (they can be changed easily)

2.) They are cost-effective

3.) They expand the tax base

4.) Targets certain products (cigarettes and more)

**Cons of indirect taxes:**

1.) They increase the price of the product (can lead to inflation)

2.) These taxes are regressive taxes (can increase the gap in income distribution in your economy)

3.) Government revenues are less certain

4.) May increase tax evasion

## Fiscal Policy

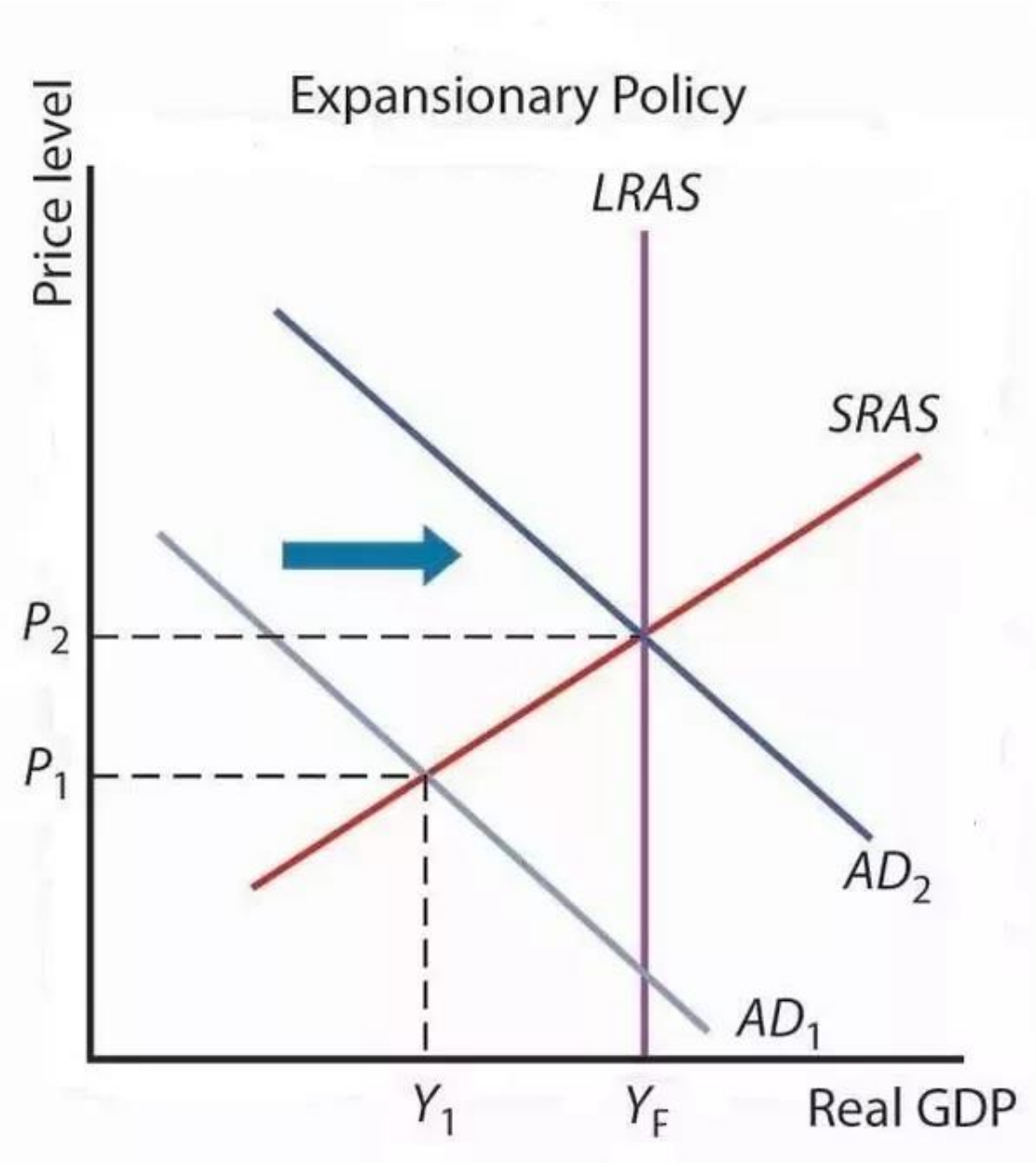
**Fiscal Policy:** A change in government expenditure or taxes in the economy to influence AD to decrease or increase (ultimately used to achieve macro-objectives like increasing GDP or decreasing inflation rate).

### Types of fiscal policy:

1.) Expansionary – To increase AD. This can be done by either increasing government expenditure or decreasing taxes in the economy.

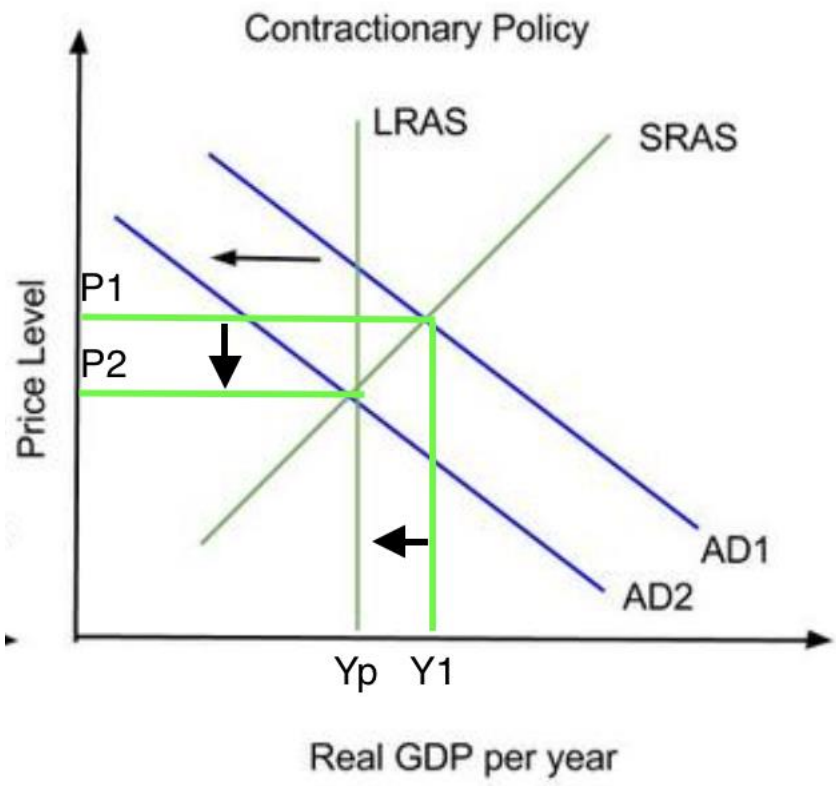
2.) Contractionary – To decrease AD. This can be done by decreasing government expenditure (usually done by selling government bonds) or increasing the taxes in the economy.

### Expansionary fiscal policy diagram:




The AD has increased from AD1 to AD2 due to the use of an expansionary fiscal policy, accordingly the RGDP of the economy increases from Y1 to the maximum potential RGDP of the economy Yf with respect to the long-run aggregate supply. At Y1 and P1 you are in a recession. Yf is the potential output that can be achieved in the long term with fixed resources in your economy. Accordingly, inflation increases from P1 to P2, but RGDP increases as well. As a result, unemployment rates decrease as well.

**<u>Contractionary fiscal policy diagram:</u>**




The AD decreases from AD1 to AD2 in this diagram as a contractionary fiscal policy was implemented. As a result, the RGDP decreases from Y1 to YP, while the inflation levels decrease from Y1 to Y2. This policy was implemented because as Y1>Yp, inflation levels in the economy are booming hence the government needs to control it. When Y1 is greater than Yp, the economy has over-exploited its resources to achieve a level of output greater than the potential. Additionally, unemployment increases in the economy.

**<u>Additional benefits of fiscal policies:</u>**

1.) When you increase direct taxes (in a contractionary fiscal policy), you take more from the rich and less from the poor which decreases the wage gap.

2.) When you tax the rich more (in a contractionary fiscal policy), the poor will spend money which adds to the circular flow of income.

3.) When you decrease taxes (in an expansionary fiscal policy), you encourage producers to invest in their company more in areas like research and development which will lead to more efficient production in your economy in the long term.

**<u>Problems with fiscal policies:</u>**

1.) It is hard for governments to determine how much to increase/decrease expenditure to make the preffered impact.

2.) During expansionary fiscal policy, as the government spends more money it leads to the crowding out effect. As the governments take more loans from the central bank, it reduces the amount of money that banks can loan out to individual consumers in the economy. Hence, as the government is spending more on the economy, individuals and companies end up spending less as they cannot take loans from the banks. Also, companies and the government will be in competition to take loans from banks, accordingly, banks will increase interest rates making it more expensive to borrow money which is also part of the crowding out effect.

3.) Increasing taxes on income or firms reduces incentive to work and incentive to invest in their companies during contractionary fiscal policies.

4.) Expansionary policy will cause an increase in inflation.

**Fiscal policy global rules:**

1.) A cap on public expenditure for a government is generalized

2.) Your growth in tax revenues cannot increase your growth in GDP

3.) You can only take a debt in proportion to your GDP

4.) Public revenues and public expenditures should be balanced

5.) The public sector should only borrow money to spend on capital because it will give returns in the future

**Crowding out effect:** When the government takes loans from the banks, individual consumers and companies will find it harder to take loans from the banks. Hence, as the government is spending more, consumer expenditure and company investments decrease in the economy simultaneously. This is due to the fact that companies and the government are in competition to take loans from the banks, thus the bank will increase the interest rates leading to less borrowing from consumers and some companies in the economy.

## Monetary Policy

**Monetary Policy:** To influence AD with the use of general interest rates of the economy. Monetary policies are controlled and set by the central bank in the economy.

**Interest rates:** The cost of borrowing money

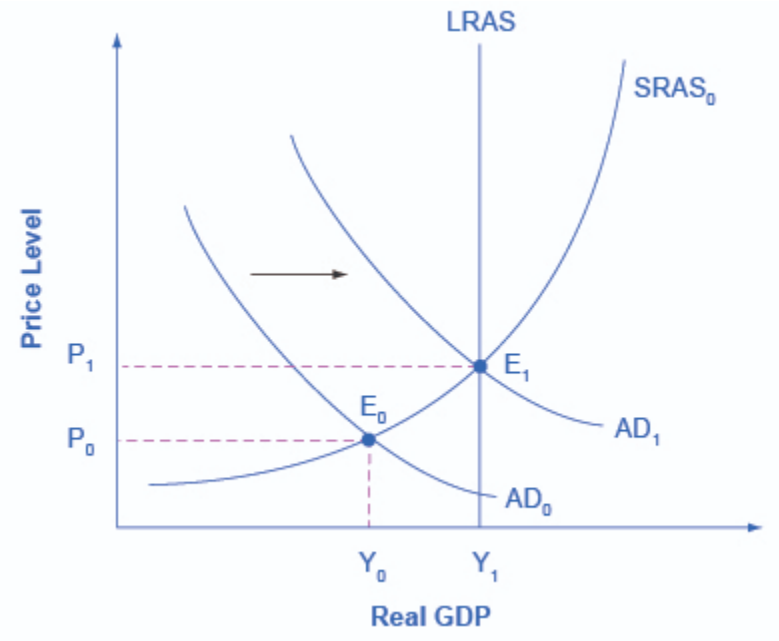
Minimum lending rate: It is the rate that the central banks sets for commercial banks when they lend money to others.

**<u>The two types of Monetary policies:</u>**

1.) Expansionary – When you increase AD in the economy by decreasing the interest rates in your economy or increasing the money supply in the economy by printing more money (not suggested).

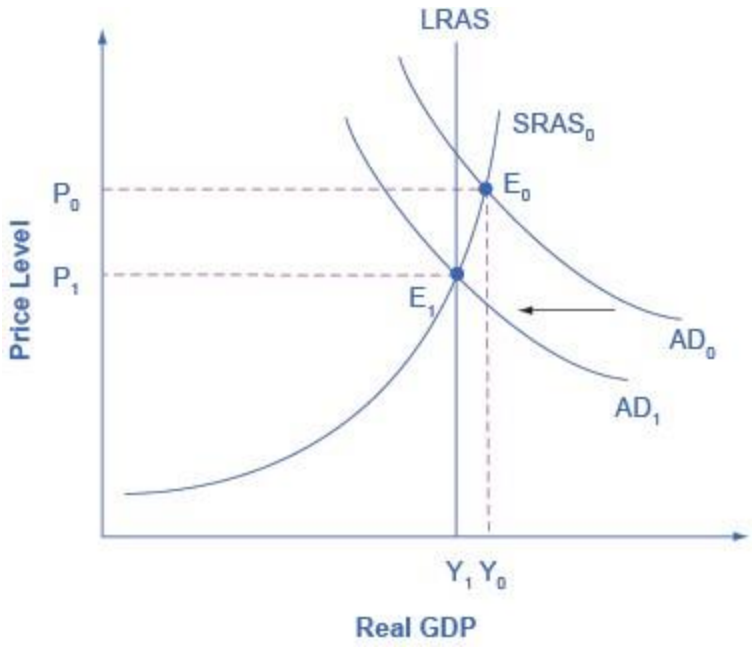
2.) Contractionary – To decrease AD by increasing the interest rates in the economy or decreasing the money supply.

**<u>Expansionary monetary policy diagram:</u>**




AD increases from AD0 to AD1 by implementing an expansionary monetary policy. Accordingly, the RGDP of the economy increases from Y0 to Y1, so unemployment rates decrease in the economy as well. However, inflation levels increase from P0 to P1.

**<u>Contractionary monetary policy diagram:</u>**




AD decreases from AD0 to AD1 by implementing a contractionary monetary policy. As a result, the RGDP decreases from Y0 to Y1, and unemployment levels increase. However, rising inflation levels are controlled as price levels decrease from P0 to P1.

**Advantages of monetary policies:**

1.) Flexibility -> the interest rates can be changed quickly and easily.

2.) If you increase interest rates it limits borrowing hence consumers are required to borrow with more responsibility so they can actually pay back their loans.

3.) Stimulates economic growth, decreases inflation, decreases unemployment according to the type of monetary policy you use.

**Cons of monetary policies:**

1.) These policies can take time for the effect to carry out and show results.

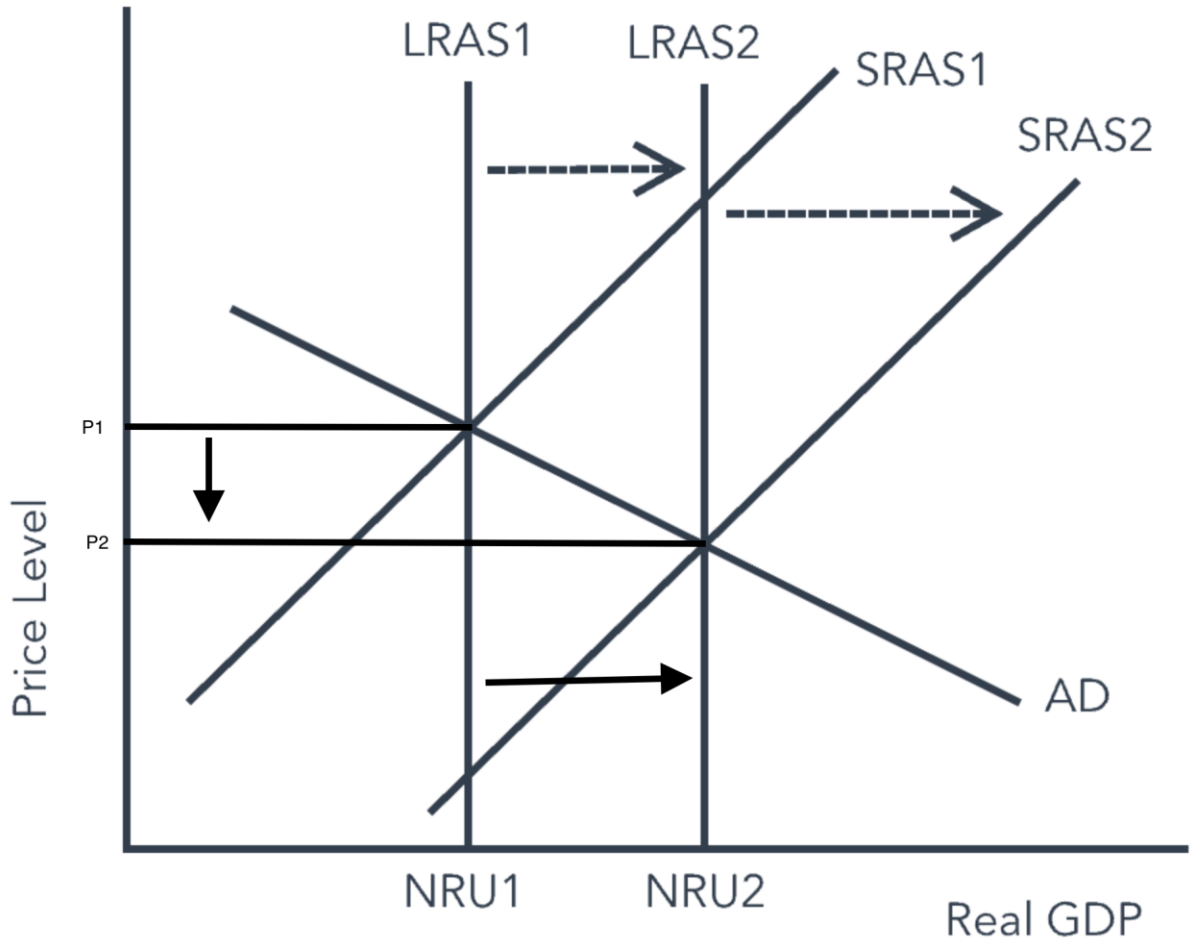
2.) If you increase the money supply of the economy by printing money it can deprecate the value of the currency and can be very dangerous in terms of rising inflation.

3.) If interest rates are lowered it can result in irresponsible borrowing of money in the economy where the borrowers won't be able to pay off their loans, thus in the long run the economy will be negatively affected.

## Supply-side policies

**Supply-side policies:** They will shift out aggregate supply to stimulate economic growth among fixing other macro-objectives. Supply-side policies affect the short run AS and aims to continue to work in affecting the long run AS.

**Supply-side diagram:**




As the supply-side policy was implemented, the short run AS was affected leading to a shift outward from SRAS1 to SRAS2. Additionally, this policy proceeded to affect the long run AS also leading to a shift outward from LRAS1 to LRAS2. As a result, the RGDP of the economy increased from NRU1 to NRU2, also unemployment rates decreased. To add on, price levels also decreased from P1 to P2 to control inflation.

Normally, supply-side policies are preferred in action because they control inflation and boost economic activity, whilst demand-side policies fix one macro-objective but worsen another.

**Different ways of supply-side policies respective to taxes:**

1.) Tax reductions in income tax -> Employees will be more motivated to work leading to more productivity.

2.) Decreasing corporate tax -> influences investments in companies which would boost productivity.

3.) Reduction in tax for profit -> Encourages firms to produce more to gain more money.

**Different ways of supply-side policies respective to subsidies:**

1.) Give subsidies to startups.

2.) Subsidize firms to invest in research and development.

3.) Investment in training labor in your economy to produce skilled labor.

4.) Subsidies given to food production like agricultural sector.

**Competition policy:** Governments make regulations to stop monopolies from forming to induce high competitiveness in the markets. Privatization -> The government sells their shares to different private sector firms so there is competition in that market.

**Trade barriers:** When countries try to restrict imports by inducing regulations like tariffs.

**Tariffs:** Taxes on imports.

**Different ways of supply-side policies respective to regulations:**

1.) Removing trade boundaries -> When more company's products enter the domestic market from other countries it encourages domestic production to increase efficiency, so they increase AS.

2.) Labor market reforms -> To remove labor unions to decrease the costs of production and increase efficiency. -> Decreasing welfare benefits like unemployment benefits to encourage work. -> Allow firms to pay low wages to workers to decrease the costs of production.

3.) Regulations -> You can impose regulations to protect a small start-up, you can have unemployment rights, protection of the environment like reducing greenhouse gas emissions. You can restrict advertisements of demerit goods like cigarettes.

4.) Deregulations -> Removing regulations to make it easier for companies to set-up and operate, hence increasing efficiency.